

IBM BOB HACKATHON 2026

HarborLens

Explainable port congestion planning for the next 72 hours

Port operations copilot | AI track

Team: HarborLens Crew

Synthetic demo data. Human approval remains required.

Congestion appears after the useful action window

Port supervisors coordinate vessel arrivals, berth capacity, crane availability, and priority cargo across disconnected schedules and spreadsheets.

Fragmented inputs

Schedules show events, but not the operational trade-offs between vessels.

Cascading delays

A late vessel or reduced berth capacity can create queue pressure across the shift.

Priority cargo

Cold-chain and connection-sensitive loads require a defensible response before they wait offshore.

Challenge context: 2021 LA/Long Beach backlog exceeded 100 ships, with an estimated \$10B supply-chain cost.

Source: Industry Problem Statements 2026, Logistics and Ports L1

A transparent recommendation engine for the supervisor's shift

- 1

Ingest snapshot

Vessel schedules, berth capacity, cargo urgency, and disruption settings.
- 2

Score risk

Deterministic scoring exposes utilization, dwell, urgency, and scenario drivers.
- 3

Plan response

A compatible berth sequence and crane allocation create a 72-hour operating plan.
- 4

Brief with Bob

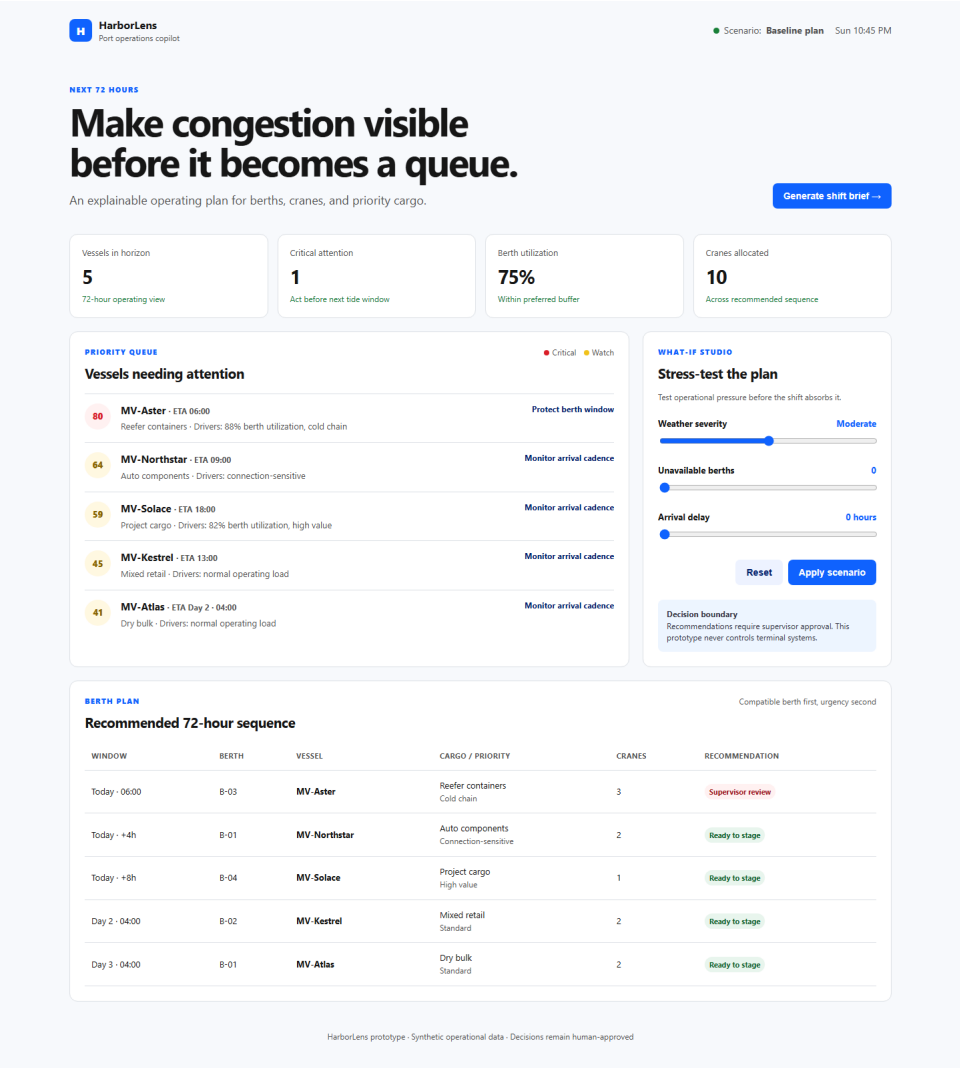
IBM Bob reviews grounded context and drafts an auditable BLUF shift brief.

The prototype never controls terminal systems. Supervisors review recommendations before any action.

Risk queue, what-if simulation, and grounded shift brief

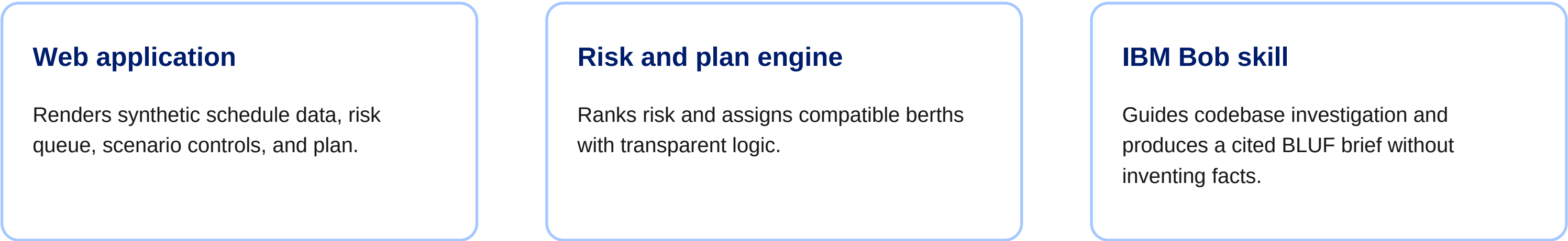
Evidence in code

- Explainable risk drivers
- Berth and crane recommendation
- Weather, capacity, and ETA scenario
- Bob-ready structured briefing context



Deterministic planning plus IBM Bob investigation

HarborLens keeps calculations and the narrative aligned by using the same computed plan for the dashboard and Bob context.



Next step: connect authenticated terminal, AIS, weather, and labor feeds while preserving human approval and plan audit history.